Computer Vision

HW 1B: Color Edge Detection

Tyler Mitts

16 February 2026

Introduction

Edge detection has many uses for image processing and is crucial for extracting features and information from images. This paper will explore some different options for edge detection, those options being gradient magnitude from grayscale image and computing 2D color structure tensor using RGB image.

Experiments and Results

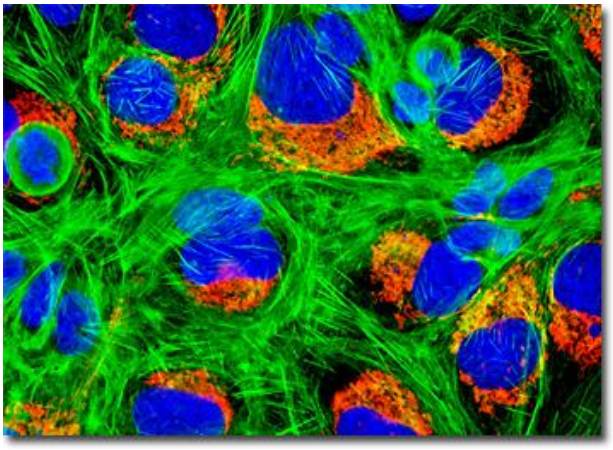


Figure 1. Cell test image.

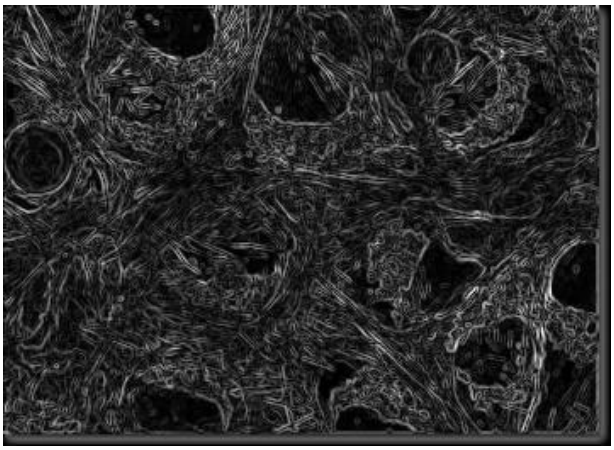


Figure 2. Gray scale magnitude edge detection.

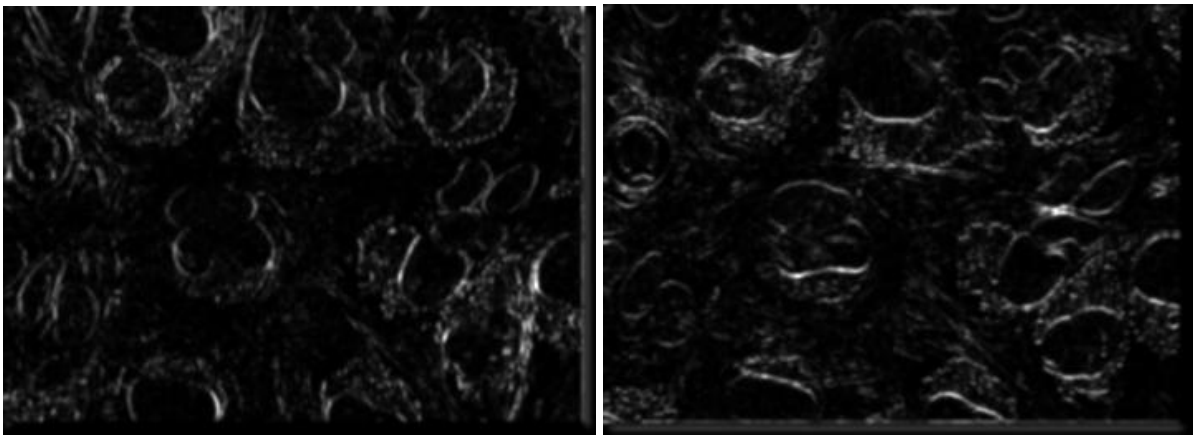


Figure 3. X and Y image gradient with gaussian blur with a sigma of 1.

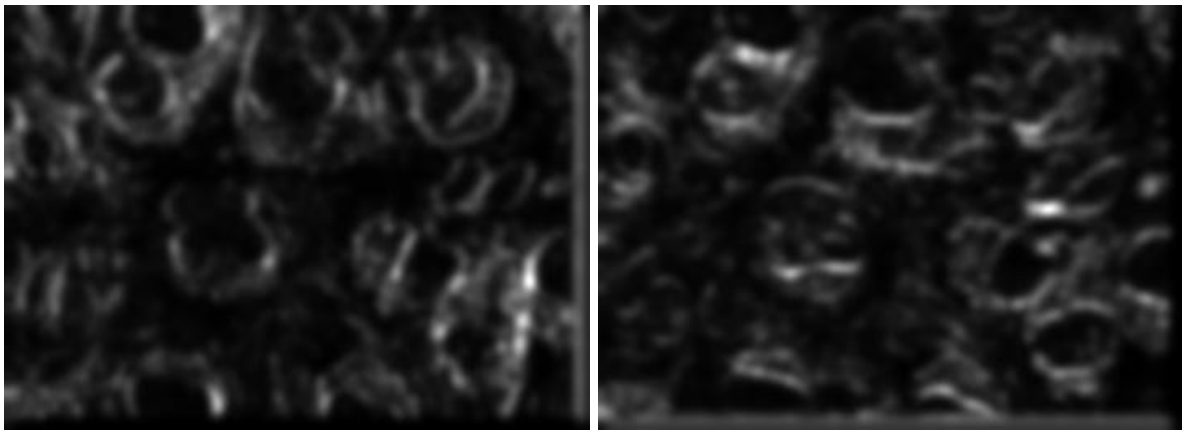


Figure 4. X and Y image gradient with gaussian blur with a sigma of 3.

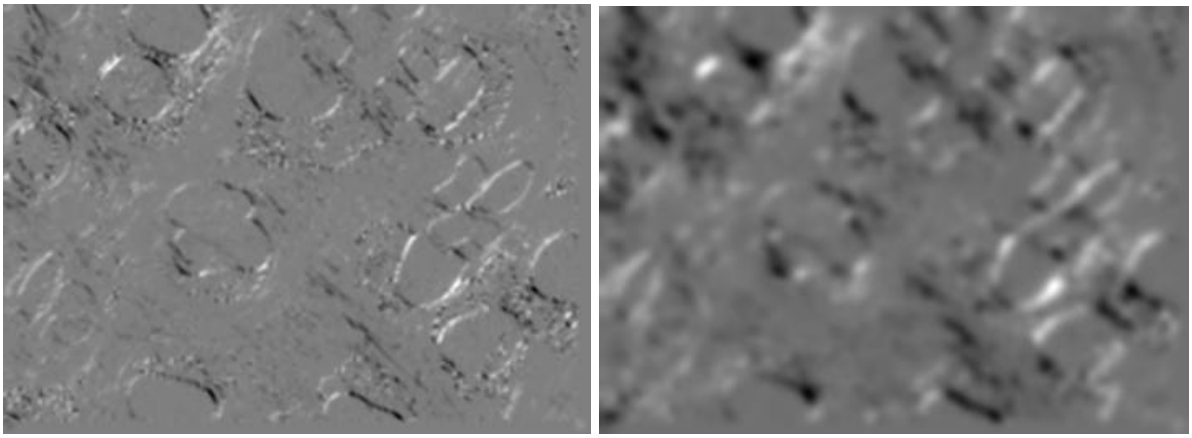


Figure 5. XY image gradient with sigmas of 1 and 3.

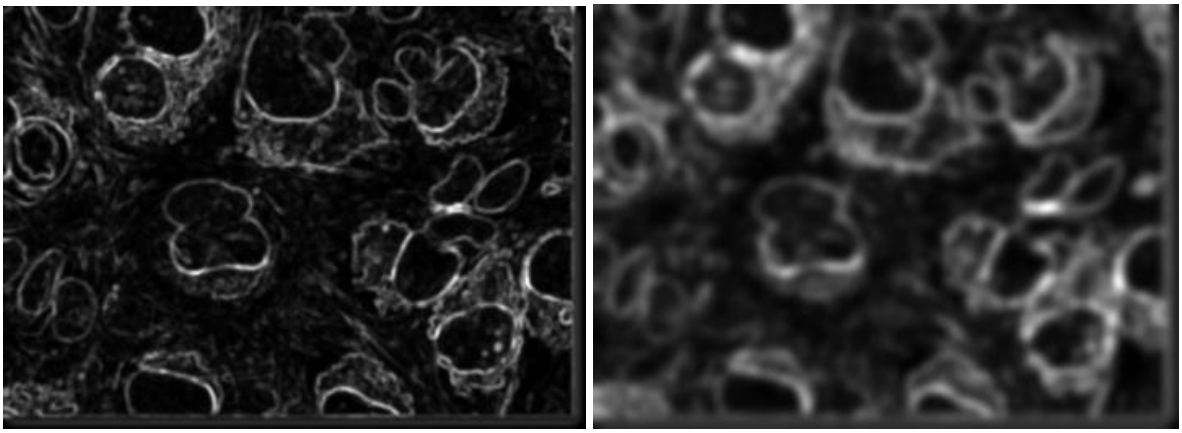


Figure 6. Trace image with gaussian blur with sigmas of 1 and 3.

Discussion

From figures 2 and 5, there is a distinct difference between the edge detection methods of grayscale magnitude and using the 2D color structure tensor. The tensor has much clearer edges and much less noise compared to grayscale. This extra noise in the gray scale can be attributed to the loss of information that happens when the image is converted into grayscale. The tensor is able to make use of the information from all color channels and combine them for a much cleaner edge detection.

When comparing the gaussian sigma values for figures 3, 4, 5, and 6, it shows that the images using sigma=1, have much more defined edges, but also capture a bit more of the extra noise found in between the cells. With the images using sigma=3, some information is lost in where exactly the edges lie, but some of the noise between the cells gets smoothed out into the black background, reducing it slightly. Depending on the use cases, both could probably be useful, depending on the criteria needed. I think that if a threshold was implemented after computing the edge detection, it could further yield better results by getting rid of more of the noise and better isolating the cells.

Works Cited

"Image Gradients." *OpenCV*, docs.opencv.org/4.x/d5/d0f/tutorial_py_gradients.html. Accessed 17 Feb. 2026.